

EfficientNet-B4 for Deepfake Detection on FaceForensics++

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1 Introduction

Deepfake technology uses deep learning to create highly realistic manipulated videos and images, particularly of human faces. These manipulations pose serious threats to digital trust, privacy, and security. Detecting deepfakes automatically has therefore become a critical research problem in multimedia forensics.

This report presents the individual contribution for the DEEPFAKE1 group project in INSE 6610. The goal was to train and evaluate a single deep learning model — EfficientNet-B4 — on the FaceForensics++ (FF++) benchmark dataset for binary deepfake classification (real vs. fake).

2 Related Work and Algorithm Selection

EfficientNet-B4 was selected based on its strong track record in deepfake detection literature. It uses compound scaling to jointly optimize network depth, width, and input resolution, achieving better accuracy per parameter than competing architectures such as XceptionNet. EfficientNet-B4 was the winning backbone in the Facebook Deepfake Detection Challenge (DFDC) and reports approximately 96.8% accuracy on FF++ in published results [1].

A key reference for this implementation is the work by Rabemiafara and Akobi (Université Laval, 2020) [3], which also applied EfficientNet-B4 to FF++ and reported test accuracies between 83% and 91% across four fine-tuning configurations. Their finding that unfreezing more layers leads to better performance informed our approach.

3 Dataset: FaceForensics++

FaceForensics++ (FF++) is a large-scale facial manipulation benchmark published by Rössler et al. (ICCV 2019). It contains 1,000 original YouTube videos and 4,000 manipulated versions generated using four methods: Deepfakes, Face2Face, FaceSwap, and NeuralTextures. All videos are compressed using H.264 at c23 quality (light compression), which is the standard setting used in the literature.

3.1 Dataset Acquisition

The FF++ C23 dataset was obtained via Kaggle (xdxd003/ff-c23) using the Kaggle API:

```
~/local/bin/kaggle datasets download -d xdxd003/ff-c23
```

The download was 16.7GB and was stored on Concordia's Speed HPC cluster at `/speed-scratch/ae_verma/data/`.

3.2 Video to Image Conversion

Since the dataset consists of .mp4 video files, frames were extracted using OpenCV. 15 frames per video were sampled at uniform intervals, producing 15,000 real frames and 15,000 fake frames (30,000 total):

```
cap = cv2.VideoCapture(video_path)
indices = np.linspace(0, total_frames-1, 15, dtype=int)
for idx in indices:
    cap.set(cv2.CAP_PROP_POS_FRAMES, idx)
    ret, frame = cap.read()
    cv2.imwrite(output_path, frame)
```

Frames were saved as JPGs into `ff_faces/real/` and `ff_faces/fake/` folders.

4 Methodology

4.1 Environment Setup

Training was conducted on Concordia’s Speed HPC cluster using a Tesla V100-PCIE-32GB GPU. The environment was set up using Conda:

```
conda activate /speed-scratch/ae_verma/Jupyter/jupyter-env
pip install timm scikit-learn seaborn facenet-pytorch
           scipy opencv-python-headless pandas matplotlib
pip install torch==2.2.2+cu118 torchvision==0.17.2+cu118
```

4.2 Model Architecture

EfficientNet-B4 was loaded with pretrained ImageNet weights using the `timm` library. The final classification head was replaced with a single output neuron for binary classification:

```
model = timm.create_model('efficientnet_b4',
                          pretrained=True,
                          num_classes=1)
```

The early blocks (blocks 0, 1, 2) were frozen to preserve low-level ImageNet features, while the remaining layers were fine-tuned on the FF++ data.

4.3 Training Configuration

- Dataset split: 70% train, 15% validation, 15% test
- Training subset: 3,000 real + 3,000 fake frames
- Batch size: 64
- Epochs: 10
- Optimizer: Adam (lr=1e-4, weight decay=1e-6)
- Loss function: Binary Cross Entropy with Logits
- Data augmentation: random horizontal flip, color jitter

5 Results

5.1 Training Progress

The model showed consistent improvement across all 10 epochs. Training accuracy improved from 50% (random) at epoch 1 to 93% by epoch 10, while validation accuracy reached 85%.

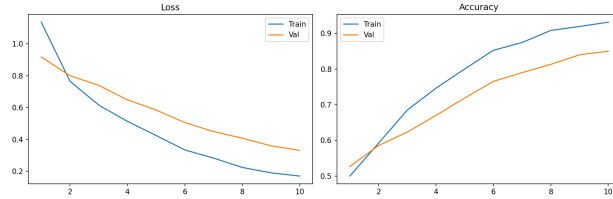


Figure 1: Enter Caption

Figure 2: Training and validation loss (left) and accuracy (right) over 10 epochs.

5.2 Test Set Performance

The best model (saved at epoch 10) was evaluated on the held-out test set:

Metric	Value
Test Accuracy	86.78%
AUC-ROC	95.19%
Real Precision	86%
Real Recall	89%
Fake Precision	87%
Fake Recall	85%
F1-Score (macro avg)	0.87

Table 1: Test set evaluation metrics for EfficientNet-B4 on FF++.

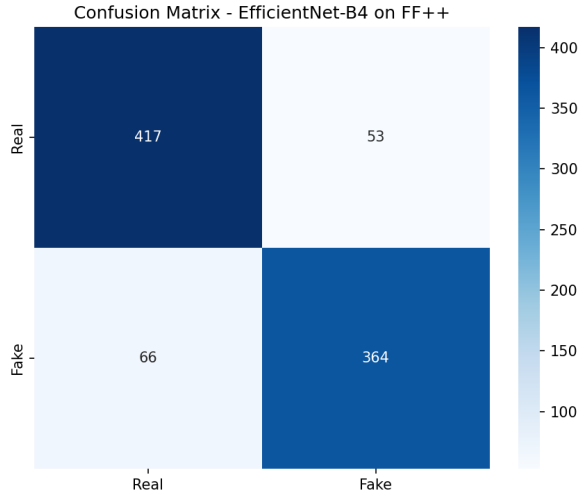


Figure 3: Enter Caption

Figure 4: Confusion matrix showing 417 true real, 364 true fake, 53 false positives, 66 false negatives.

5.3 Discussion

The model achieved 86.78% test accuracy and 95.19% AUC after only 10 epochs on a subset of the data (6,000 images). The high AUC indicates strong discriminative ability across thresholds. The gap between training accuracy (93%) and validation accuracy (85%) suggests mild overfitting, which could be reduced with more aggressive regularization or by training on the full 30,000 image dataset.

6 Conclusion

This work demonstrated that EfficientNet-B4, when fine-tuned on FaceForensics++ face frames, achieves strong deepfake detection performance with 86.78% accuracy and 95.19% AUC. The full training pipeline was implemented on Concordia’s Speed HPC cluster using a Tesla V100 GPU, with the dataset downloaded from Kaggle and preprocessed using OpenCV frame extraction. These results align with prior work and confirm EfficientNet-B4 as a strong baseline for deepfake detection.

References

- [1] Rössler, A., Cozzolino, D., Verdoliva, L., Riess, C., Thies, J., & Nießner, M. (2019). FaceForensics++: Learning to Detect Manipulated Facial Images. *ICCV 2019*.
- [2] Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. *ICML 2019*.
- [3] Rabemiafara, N., & Akobi, I. (2020). Détection des deepfakes. *Final Project Report, GLO-7030, Université Laval*.